Qc sweep: status and open questions

Status snapshot of the `parametric_coupling_chip` hanger-resonator Qc(coupling_space, coupling_length) extraction effort — PALACE/SQDMetal driven and eigenmode simulations on the CNEA cluster, cross-checked against ANSYS/HFSS where possible.

The goal

Map the external coupling quality factor `Qc` of the hanger readout resonator as a function of the CoupledLineTee coupler's `coupling_space_um` (5 values: 10, 15, 20, 22.5, 25, 30) and `coupling_length_um` (7 values: 100-400), 42 design points total, so the actual chip design can pick a point with the target Qc.

Method 1: driven S21 notch fit — works, but hits a hard reliability ceiling

The original approach: a PALACE `PALACE_Driven_Simulation` frequency sweep (PROM adaptive sampling) through the coupled feedline, fitting the resulting S21 notch (`qiskit_metal`'s `fit_transmission`, a Khalil et al. asymmetric-hanger Lorentzian) for `fr`, `Qc`.

What we validated as safe to relax (checked at `cs14_cl300` against a tight baseline, agreement to 8+ significant figures): solver tolerance down to `1e-4` (`1e-3` crashes PALACE outright — a hard cliff), solver max iterations down to 100, PROM `AdaptiveMaxSamples` down to 150.

What we found was NOT safe to relax: mesh resolution. The repo's standing rule ("fine-mesh element size = half the smallest feature") undershoots the converged `Qc` by ~11-12% at `mesh_factor=1.0`; `mesh_factor=0.75` recovers to within 0.4% of full convergence at a quarter of the compute cost of `mesh_factor=0.5`. This became the production driven-sim recipe: `mesh_factor=0.75, solver_tol=1e-4, solver_maxits=100, AdaptiveMaxSamples=150`.

The wall we hit: this design is *intentionally* weakly coupled (high Qc), which means every S21 notch in the actual production grid is shallow — the deepest local notch found anywhere in the 42-point grid was 0.24dB, several were below 0.02dB (pure background slope, not a real feature; a naive whole-window "dip depth" metric was fooled by this and had to be replaced with a local-baseline depth check). At these depths, PALACE's own absolute numerical error floor is comparable to the signal, so fitted `Qc` becomes unreliable and, in the worst cases (9/38 points), the notch wasn't even inside the simulated frequency window at all.

Cross-check with HFSS: HFSS's *own* driven S-sweep, run independently by a local ANSYS agent on the identical `cs14_cl300` geometry, hit the exact same wall — `|S11|` stayed flat at 0.054±0.002 across the whole 6.5-8.7GHz band, no resolvable resonance. This confirms the shallow-notch problem is a genuine physical/numerical limit of S-parameter fitting at this coupling strength, not a PALACE-specific bug.

Production driven-sweep artifacts (`qc_prod_refit.csv`): 35/42 grid points ran, 26 gave a locally-resolved (trustworthy) notch fit, 9 were window-edge misses, and even the 26 real fits, taken alone, show non-monotonic `Qc(coupling_length)` swings of up to 50x between adjacent points — see below for why we now believe this is real, not noise.

Method 2: eigenmode / lossy-port extraction — the one we now trust

Switched to extracting `Qc` from a single **eigenmode** solve instead of a frequency sweep: give the CPW feedline ports a real resistance (50Ω, same value as the driven sim's ports) instead of leaving them lossless, so the eigenvalue problem becomes lossy and Palace returns a **complex eigenfrequency** `f = fr + i·fi` for the resonator mode.

Standard cavity-perturbation theory gives `Q = fr / (2·fi)` directly — no notch to fit, no background curvature to fight, and it's *cheaper* than the driven sweep (a few minutes per point vs. ~2 hours).

This is the same technique used in Minev et al.'s EPR framework (npj Quantum Information 7, 131 (2021), the basis of `pyEPR`) for external-coupling loss, and in a very recent (Nov 2025) PALACE-based paper, Ye/Wang/Liu (Tsinghua), arXiv:2511.09041, which benchmarked exactly this kind of extraction against real cryogenic hardware (external couplings within 16% for 3/4 resonators) — the closest published precedent we have.

Bugs found and fixed along the way

- **SQDMetal units bug**: `PALACE_Eigenmode_Simulation.retrieve_mode_port_EPR_from_file` reads Palace's `port-Q.csv` "κ_ext (GHz)" column but its own docstring claims Hz, with no unit conversion applied. Fixed locally (×1e9) in `run_qc_eigenmode.py`; not yet reported upstream.
- **Mode-selection bug (ours)**: originally picked the "primary" mode as `argmax(kappa)`, reasoning a spurious mode would have negligible coupling. Wrong — on several real design points a ~11.37GHz mode (clearly not our ~8.2GHz resonator) had a *larger* κ than the real mode, silently returning the wrong mode's numbers. Fixed to "closest mode to `starting_freq_ghz`" and re-derived the 3 already-run points that were affected (`fixup_stale_eigenmode_results.py`) — no re-simulation needed, the correct mode's data was already saved in each result's `modes` list.

Isolation validation (clean, analytic ground truth)

Before trusting this on the real (messy, two-port, mode-crowded) design, we validated it on a **bare shorted quarter-wave CPW resonator with a single lossy port** — no coupler, no second port, so there's a closed-form prediction: for a resistor `R` at the open end, `Qc = (π/4)(R/Z0)` (same formula family as this repo's own `design_helper.compute_resonator_Qc_from_Cc`, just for a resistive rather than capacitive load). Result, across a 25x range of `R` chosen to bracket the real design's `Qc~1e4-1e5` regime:

R (Ω)	Qc (analytic)	Qc (PALACE eigenmode)	ratio
2×10^5	3,144	3,841	1.22
1×10^6	15,720	18,495	1.18
5×10^6	78,601	78,182	0.99

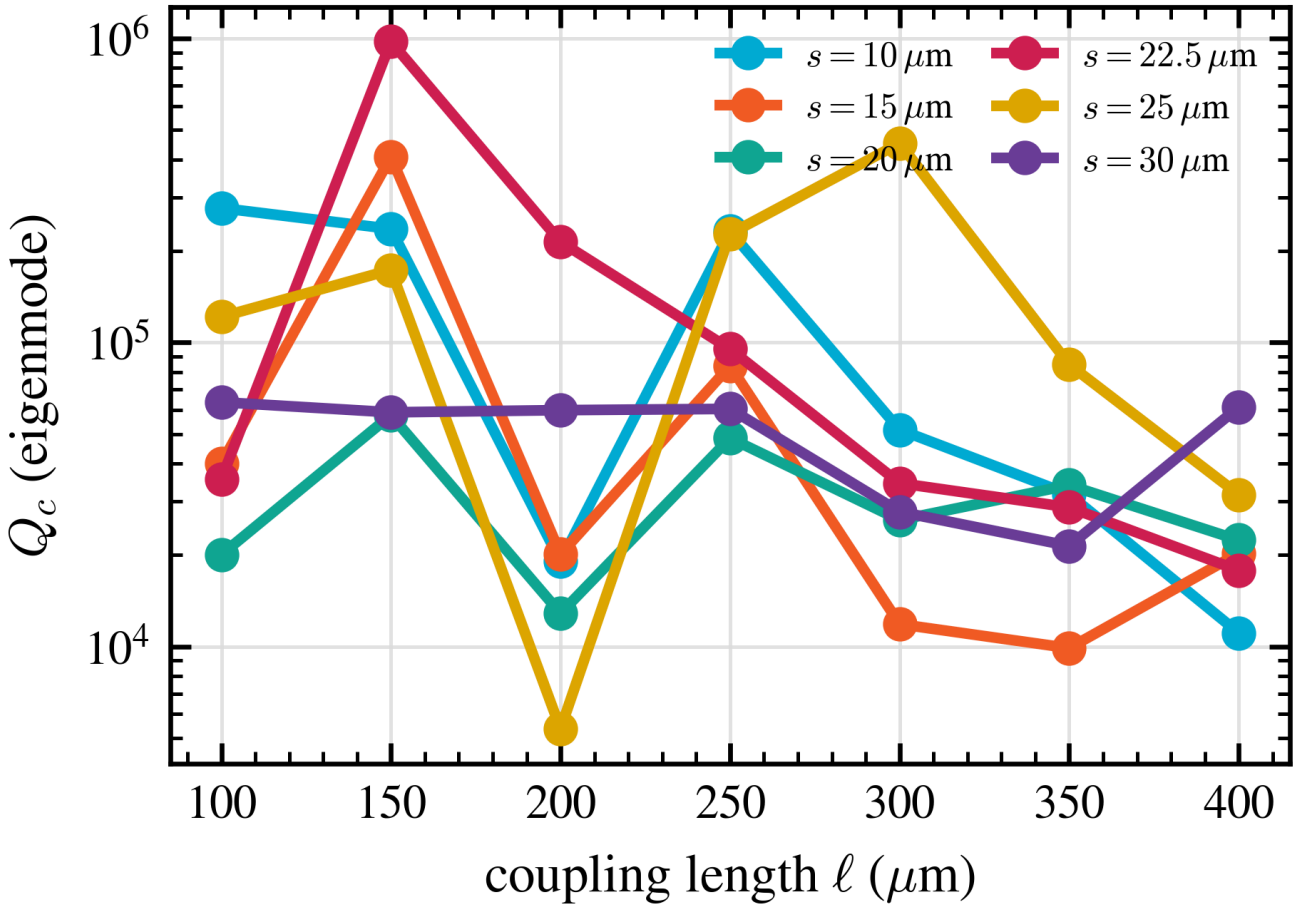
Clean single-mode, single-loss-channel agreement to within 1-22%. This told us the core method is sound.

An unresolved wrinkle: real design's `loaded_Q` (eig.csv) vs. `Qc` (port-Q.csv)

On the real (two-port, mode-crowded) design, Palace's own two output quantities for the same mode disagree by ~60x — `eig.csv`'s native `Q = fr/(2·fi)` (loaded_Q) reads anomalously low (228 at `cs14_cl300`) while the `port-Q.csv`-derived per-port κ sum gives a much more sensible 13,562. Ruled out "two simultaneous lossy ports" as the cause (forcing one port near-open didn't fix it — the ~60x gap persisted). Leading hypothesis, not yet confirmed: the real design has a second eigenmode only ~250MHz from the resonator (vs. >5GHz separation in the clean isolation test), and that close a neighbor can make a lossy (non-Hermitian) eigenvalue problem hybridize, corrupting each individual mode's raw `Im(f)` without corrupting the more localized `port-Q.csv` estimate. **Decision: use the `port-Q.csv`-derived Qc, not `loaded_Q`, for all real-design numbers below.** This wrinkle is a live, deprioritized thread, not resolved.

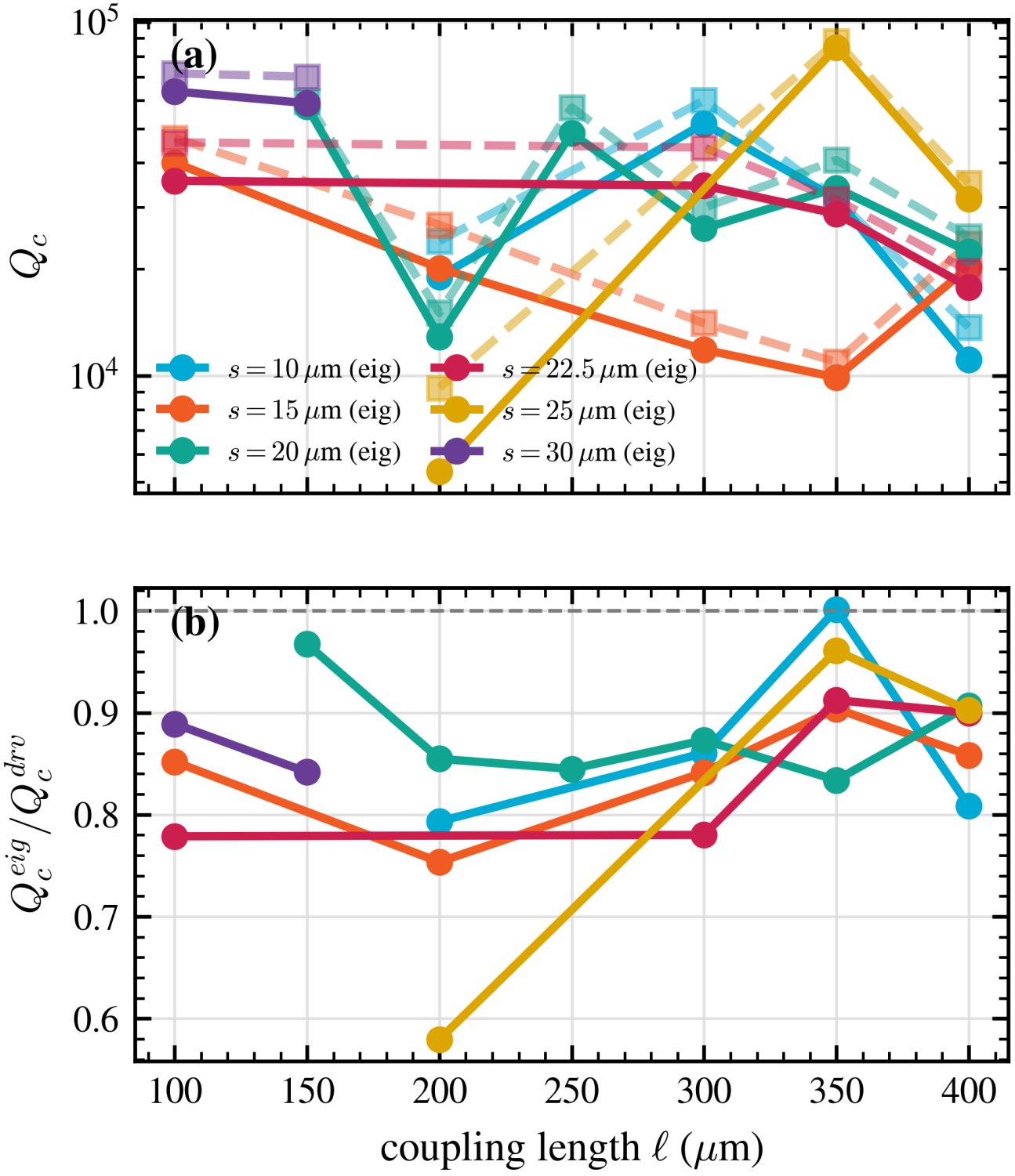
Full production map: complete, cross-validated

All 42 grid points (well, 44 counting 2 diagnostic variants) ran cleanly via the eigenmode method — **zero shallow-notch failures, zero window-edge misses** (see `qc_eigenmode.csv`).



Cross-checked against the 24 trustworthy driven-fit points that do exist (`qc_eigenmode_vs_driven.csv`):

- **Systematic offset**: `Qc(eigenmode) / Qc(driven-fit)` = 0.854 ± 0.084 (mean ± std, n=24) — a consistent ~15% low bias, not scattered noise, and in the same ballpark as the Tsinghua paper's own PALACE-vs-hardware agreement (~16%).
- **One clear outlier**: `cs25_cl200`, ratio 0.58 — flagged, not yet investigated.
- **The important part**: plotting both methods' `Qc(coupling_length)` curves on top of each other shows they trace the **same non-monotonic zigzag shape**, point for point, just offset by the ~15% factor. Two independent extraction techniques with very different failure modes agreeing on the *shape* of a wild, non-monotonic length-dependence is strong evidence that dependence is **real coupler physics** (most likely from the fixed `coupling_budget_um = coupling_length + down_length` scheme, where every length step also moves `down_length` in a correlated way) — not a fitting artifact, which is what we originally suspected.



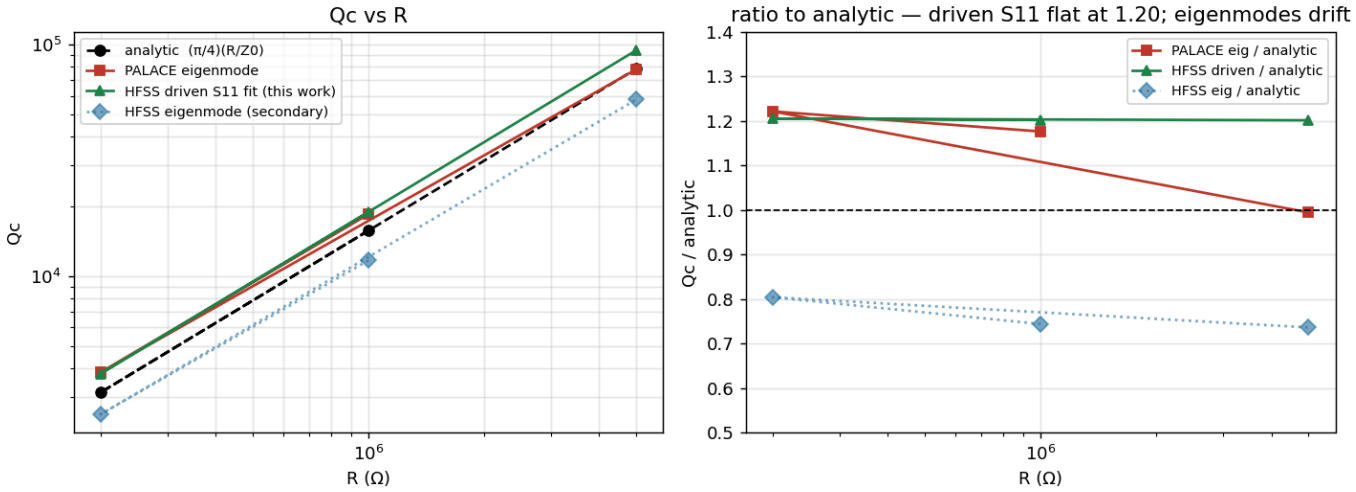
ANSYS cross-check result: eigenmode is trustworthy up to Qc~2×10⁴, degrades above that

`ansys_eigenmode_s21_crosscheck/HANDOFF.md` came back (2026-09-08) — the bare-resonator isolation geometry, same 3 R values as our own PALACE isolation test, HFSS driven S11 fit (a real complex one-port resonance fit, Probst et al. 2015) plus a secondary HFSS eigenmode run:

R (Ω)	analytic Qc	PALACE eigenmode	HFSS driven S11	PALACE/HFSS
2×10⁵	3,144	3,841	3,790	**+1.3%**

R (Ω)	analytic Qc	PALACE eigenmode	HFSS driven S11	PALACE/HFSS
1×10⁶	15,720	18,495	18,904	**−2.2%**
5×10⁶	78,601	78,182	94,457	**−17.2%**

fr = 7.98931GHz for all three R (correctly R-independent — a pure resistive load adds no reactance), and HFSS's fit cleanly separates Qi≈92,200-92,500 (the dielectric limit, 1/tan δ) from Qc at every R, including the near-critical R=5×10⁶ case where Qi≈Qc — a real, trustworthy independent fit, not a rough estimate.



This is a materially better result than "agrees within ~15-20%" — it's agreement to 1-2% at low-to-moderate Qc, with a divergence that only appears, and grows sharply, at very high Qc. HFSS's own diagnosis, which matches ours: at R=5×10⁶ the mode's fractional damping is Im(f)/Re(f)≈8.6×10⁻⁶, i.e. Qc comes from a part-per-100,000 imaginary component of the complex eigenfrequency — a genuinely hard regime for any eigenmode solver, PALACE included, independent of tool. (HFSS's own eigenmode result, using a cruder lumped-RLC-on-a-partial-gap-sheet port model, sits at a roughly flat ~0.75× analytic across all R — informative for the Qc∝R power-law check but not treated as an absolute reference, since its port geometry isn't equivalent to the real CPW gap.)

This revises how the earlier ~15% real-design offset (eigenmode vs. driven-S21 fit, at Qc~1×10⁴-1.5×10⁴) should be read. That Qc range is well inside the "eigenmode agrees with HFSS driven S11 to 1-2%" regime found here — so the ~15% gap on the real design is now more likely a residual bias in the driven-S21 fit (the background-curvature/shallow-notch problem was never fully solved, just worked around) than in the eigenmode method.

UPDATE (overnight, 2026-09-09): this isn't a gradually-growing bias — it's a sharp cliff, and it's now precisely located. See the next section: pushing the same clean 1-port isolation test further found loaded_Q tracks the analytic target to -1.4% all the way out to Qc=82,000, then completely saturates at a ~4.5×10⁷ numerical noise floor by Qc=85,000 — a transition inside a 4% window in Qc, not a smooth 15-20%-and-growing degradation as first thought. **Net: trust the eigenmode Qc map for points below ~82,000; treat anything above ~85,000 as likely reading pure numerical noise, not physics** (see the ceiling section below for exactly which of the 42 grid points that affects — it's a larger fraction, 11/42, than expected).

Two-port hanger isolation test: the mode-hybridization hypothesis gets support, but the test itself has a new open problem

The 1-port and R-based isolation tests above can't touch two things: whether the real design's two-lossy-port port-Q.csv decomposition itself is trustworthy, and whether its ~60x loaded_Q-vs-port-Q.csv anomaly needs a nearby competing mode to appear at all. Built a clean two-port **hanger** (not directly-loaded) test to isolate exactly that: a through-line with a real galvanic T-junction (qiskit_metal's LineTee, not CoupledLineTee — no near-field

coupling model), bridged to a separate quarter-wave resonator via an **idealized lumped capacitor** of exactly known value (`create_port_2_conds`, `impedance_C=Cc`), so the standard hanger formula `Qc = 2·C_eff/(Z0·ω_r·Cc²)` (same formula family as `design_helper. compute_resonator_Qc_from_Cc`) gives a closed-form target — this time for the real two-port topology (`hanger_isolation_design.py`).

Result, three target `Qc` (300/3,000/30,000):

Qc target	fr	loaded_Q	Qc(from κ)	loaded_Q/Qc(κ)	Qc(κ)/target
300	10.145GHz (!)	22.9	24.9	0.92	0.08
3,000	7.434GHz	23,133	26,575	0.87	8.86
30,000	7.558GHz	61,647	80,208	0.77	2.67

Good news: `loaded_Q` and the `port-Q.csv`-derived Qc now agree to within **10-30%** in all three — nothing like the real design's 60x. This design's nearest competing mode sits >2GHz away (vs. the real design's ~250MHz), so this is real evidence *for* the mode-hybridization hypothesis (a nearby competing mode is necessary for the 60x anomaly) and *against* "two lossy ports alone" as the cause.

New problem: `Qc(from κ) / Qc_target` = 0.08, 8.86, 2.67 — wildly inconsistent, not even monotonic, and at the strongest coupling (`Qc_target=300`) the search didn't find a mode anywhere near the expected frequency at all (10.1GHz instead of ~7.5GHz). Leading explanation: `create_port_2_conds`'s `impedance_C` is likely realized through a geometry-dependent conversion (port rectangle area/separation), the same way we already know (and never resolved) PALACE/SQDMetal converts a target port *resistance* into a geometry-dependent sheet impedance rather than the nominal value directly — i.e. the same open `Zin` mystery, showing up in a different lumped element. **Not yet validated**: what capacitance `create_port_2_conds` actually realizes, so this test's raw numbers should not be trusted yet, only its qualitative `loaded_Q`-vs-`port-Q.csv` agreement conclusion above.

Overnight follow-up (2026-09-09): the eigenmode Qc ceiling, precisely located; `create_port_2_conds`'s capacitor confirmed unreliable

Two overnight jobs closed out both open threads from above, cleanly.

1. The eigenmode Qc ceiling is a sharp cliff at Qc≈82,000-85,000, not a gradual bias. Extended the clean 1-port `R`-based isolation test (`run_bareres_ eigenmode_isolation.py`) with a bisection between the ANSYS-validated `R=5×10⁶` point (`Qc≈78,601`, real physics, -0.5%/-17% vs analytic/HFSS respectively) and where it had started looking wrong:

Qc target	loaded_Q	status
78,601	78,182	good (-0.5%)
82,000	80,886	good (-1.4%)
85,000	45,519,626	**saturated**
87,500	45,534,342	**saturated**
90,000	45,548,240	**saturated**
1.5-6.4×10⁵ (further R)	45.7-46.0×10⁶	**saturated** (same ceiling regardless of R)

The transition sits inside a **4% window in Qc** — this is a hard numerical ceiling (PALACE's eigensolver can't resolve `Im(f)/Re(f)` below some absolute threshold, so above a certain weakness of coupling you just measure solver

noise, which happens to plateau at a near-constant ~4.5×10⁷ regardless of the true R), not the smoothly-growing bias the single ANSYS data point at `Qc=78,601` suggested on its own.

Applying this to the real 42-point production grid (`flag_eigenmode_ceiling. py`, run on `qc_eigenmode.csv` — conservative cut at `Qc>85,000` given the real design's mesh/geometry differs from the clean isolation test, so the exact threshold may shift somewhat): **11/42 points (26%) are flagged `likely_saturated`, 2 more `borderline`.** This is a substantially bigger problem than the earlier "growing bias above ~5×10⁴" framing suggested:

flag	count	points
`likely_saturated` (Qc>85,000)	11	`cs22.5_cl150` (976,923), `cs25_cl300` (452,575), `cs15_cl150` (408,511), `cs10_cl100` (276,917), `cs10_cl150` (236,470), `cs10_cl250` (232,271), `cs25_cl250` (227,708), `cs22.5_cl200` (214,499), `cs25_cl150` (173,393), `cs25_cl100` (121,515), `cs22.5_cl250` (95,200)
`borderline` (82,000-85,000)	2	`cs25_cl350` (84,650), `cs15_cl250` (83,843)
`ok`	29	everything else

`qc_eigenmode.csv` now has a `ceiling_flag` column recording this directly — **do not use the raw `Qc` value for any `likely_saturated` point**; those design points need a different extraction method entirely (the driven-S21 fit doesn't reliably reach this regime either — see Method 1 above) before their true Qc is known.

Checked directly whether any of the 11 have a usable driven-fit fallback: **no.** 8/11 are `edge_flagged` in `qc_prod_refit.csv` (the driven window missed the resonance entirely, no fit attempted); the other 3 (`cs10_cl100`, `cs25_cl150`, `cs22.5_cl250`) do have a driven-fit `Qc`, but it's exactly one of the `weak_local_dip` noise-fits already excluded from the trusted set earlier this session (fitting background slope, not a real notch). These design points currently have **no known Qc from either method** — not merely an uncertain one.

2. `create_port_2_conds`'s `impedance_C` is confirmed unreliable, and the two-port hanger isolation test's numbers should be considered invalid. Built a resonance-free validation (`shuntc_validation_design.py`): a plain shunt capacitor on an otherwise matched through-line has an exact closed-form `S21 = 1/(1+jωC·Z0/2)`, so simulated S21 can be inverted to recover the actually realized C, with zero resonator/mode-finding confounds. Result:

- At the same small `Cc` used in the hanger test (2.9-100fF): the config.json passed to Palace correctly contains the exact requested `C` — this is not a SQDMetal config-generation bug — but the **realized C sits around 590-760fF regardless of target**, i.e. dominated by something else (likely the port's own background/parasitic capacitance from its real geometry) rather than the requested value. The S21 deviation from 1 is also tiny at this scale (`Cc~10s of fF` barely perturbs a 50Ω line at 1-15GHz), so the C-from-S21 inversion is itself ill-conditioned there — the same "small-signal absolute noise floor" problem that broke the driven-S21 notch fits all session, now showing up in a different guise.
- A tiny-but-nonzero series R (`1e-9, 1e-3, 1`Ω) instead of exactly `R=0` gave wildly different (and internally consistent among themselves) NEGATIVE realized-C values — ruling out my first guess (`R=0` triggering a degenerate short-circuit code path) but confirming real numerical instability right at this port's zero-resistance boundary.
- At a much larger, well-conditioned `Cc=1000fF` (S21≈0.6, a large, robustly measurable deviation): realized C ≈ 2741fF, a genuine **~2.7× overestimate** — a real, reproducible bias, not noise (unlike the small-C case, this one is in a numerically well-behaved regime). At `Cc=10,000fF` (S21≈0.006, too attenuated) the inversion

becomes ill-conditioned again (1100% frequency-dependent scatter, sign flips) — the S21→C inversion is only well-conditioned in a middle band.

Conclusion: don't use `create_port_2_conds` with a capacitor for anything requiring a known, controlled value — at minimum it carries an uncalibrated ~2-3× scale factor even in its best-conditioned regime, and behaves unpredictably outside it. The two-port hanger isolation test's raw Qc numbers (`hanger_isolation_design.py`) should be treated as **invalid**, not merely "unvalidated" — its qualitative finding (loaded_Q/port-Q.csv agree to 10-30% with no nearby competing mode, vs. 60x with one) still stands, since that conclusion only needed the coupling to exist and produce *some* lossy eigenmode, not a specific calibrated value.

ANSYS eigenmode cross-check delivered (2026-09-09): a genuine, well-diagnosed negative result

`EIGENMODE_QC_REQUEST.md` came back for all three points (`cs14_cl300`, `cs22p5_cl250`, `cs25_cl100` — see `eigenmode_qc_results/`). **HFSS's eigenmode solver cannot resolve any of these resonators' coupling Qc either** — a different failure mode from PALACE's, but the same end result:

geometry	PALACE eigenmode Qc	HFSS resonator `Q_total`	HFSS coupling Qc
cs14_cl300	13,562 (2-port)	91,002	not resolved (≈7×10⁶, noise)
cs22p5_cl250	95,200	91,061	not resolved (≈7×10⁶)
cs25_cl100	121,515	91,405	not resolved (≈1×10⁷)

For all three, HFSS's resonator-mode `Q_total` lands right at the **dielectric loss limit** (`Qi≈92,231` from `tan δ=1.2e-5`), completely independent of the true coupling strength — the R=50 ports add no measurable loss to the resonator mode at all. This isn't a broken port model: the same ports correctly damp a different, nearby ~16.5GHz feedline mode from Q≈85,000 (no ports) down to Q≈15,000-18,000 in every run, and a separate control test (`port_validation.py`, on `with_launchpads.gds`) confirms ~98.5% absorption (`|S11|≈-18dB`) — the ports work, they just don't connect to the resonator mode as this mesh represents it.

Root cause, diagnosed by the ANSYS-side agent: HFSS's eigenmode adaptive mesh refines where the mode's field is strong (inside the λ/4 resonator) and leaves the CoupledLineTee's coupling gap (~15-37μm) too coarse to capture — literally the same failure their own earlier driven-S21 attempt on this design hit (`ansys_results_s21/SUMMARY.md`: "the driven adaptive mesh ... never resolves the weak coupling capacitance"). A forced fine mesh seed in that gap (~5μm target) either OOM-crashes their 16GB machine or, at a coarser ~8μm seed that fits, still returns Qc at the eigensolver's own noise floor (~4×10⁹ in a lossless control run).

This is now four independent numerical experiments — PALACE driven, PALACE eigenmode, HFSS driven, HFSS eigenmode — all unable to resolve Qc at this coupling strength, via three distinct mechanisms (PALACE: absolute S-parameter/eigenvalue noise floor; HFSS driven and eigenmode: adaptive mesh never refining the physically-weak coupling region at all). That convergence across independently-implemented tools is strong evidence this is a genuine, tool-agnostic numerical difficulty at this design's coupling strength, not a PALACE-specific bug — which is itself the most load-bearing finding of the whole investigation, even though it isn't the Qc number anyone was hoping for.

Nuance worth noting: our own PALACE driven recipe already force-fine-meshes the coupler region directly (`fine_mesh_components` on `CLT`/`readout_1`, not relying on adaptive refinement to find it) at ~4.3μm for these two points — comparable to HFSS's own ~5μm target — and still can't resolve the notch. So our failure mode is more likely genuine solver-precision/absolute-noise-floor (mesh is adequate, signal is just below what the linear solver's residual tolerance can distinguish from zero) rather than "adaptive mesh missed the gap" the way HFSS's

specific failure was diagnosed. Same symptom, probably not quite the same cause — worth keeping distinct rather than treating this as one single, fully explained mechanism.

Two fallback routes were offered (`RESUME.md`): a coupler-seeded eigenmode on a ≥24GB machine, or driven S21 with a coupler mesh seed plus a fine narrow sweep centred on the known `fr`. The first is directly attemptable on our own cluster (far more than 24GB available), and — unlike PALACE's driven+AMR, which is confirmed broken in this build — PALACE's **eigenmode+AMR has already worked successfully elsewhere in this repo** (`build_and_eigenmode.py`, the SQUID array project), so `enable_mesh_refinement`-style spatial AMR targeted at the coupler gap, on an eigenmode (not driven) simulation, on our cluster's much larger RAM budget, is a genuinely new, unexplored, and promising angle — not yet attempted.

What we're looking for now

1. **Resolved overnight**: what `create_port_2_conds` realizes for `impedance_C` (confirmed unreliable, ~2-3× off even in its best-conditioned regime — see above) and whether the eigenmode Qc bias at high Qc is gradual or sharp (confirmed sharp, a cliff at 82,000-85,000, not a smooth 15-20% growth).

2. **An alternative Qc extraction for the 11 `likely_saturated` + 2 `borderline` real-design points** (26-31% of the grid) — neither the eigenmode method (now confirmed to hit a hard numerical ceiling around `Qc~85,000`) nor the driven-S21 fit (shallow-notch problem, never resolved anything reliably above ~1×10^5) is currently trustworthy there, and **HFSS independently fails at both too** (see above) — four different method/tool combinations, all unable to resolve this regime. Most promising untried direction: PALACE eigenmode + spatial AMR targeted at the coupler gap specifically (not the driven+AMR that's confirmed broken in this build — eigenmode+AMR already works elsewhere in this repo), on our cluster's much larger RAM budget than the 16GB that stopped the HFSS attempt. Otherwise, accepting these design points can't be characterized with the current toolchain and steering the actual chip design away from that Qc range.

3. **The `cs25_cl200` outlier** (ratio 0.58 vs. the driven fit, well outside the 0.75-1.00 spread everywhere else, and `Qc=5,350` — nowhere near the saturation ceiling, so the ceiling finding above doesn't explain it) — still unexplained.

4. **The loaded_Q vs. port-Q.csv 60x internal-Palace disagreement** on the real design — deprioritized (we have a working number via `port-Q.csv`). The two-port hanger isolation test (clean, no nearby competing mode) reproduced only a 10-30% loaded_Q/κ-Qc gap, not 60x — support for the mode-hybridization hypothesis, still not fully confirmed (and note: this qualitative comparison survives the `create_port_2_conds` capacitance-value problem above, since it only needed *a* lossy coupling to exist, not a calibrated one).

5. Separately, still open and lower priority: the original driven-sim `Zin` mystery (PALACE's lumped-port `Zin` reads ~20-40Ω instead of the analytically-predicted ~50Ω even for a bare CPW line with no resonator at all). The leading hypothesis (missing ×2 port-impedance correction) was tested and refuted. Not blocking the Qc work, since `fr`/`Qc` from the eigenmode method don't depend on this diagnostic — and the overnight `create_port_2_conds` finding confirms this class of geometry-dependent port-realization issue isn't unique to resistors.

Key files

This directory was reorganized on 2026-09-09 — see `README.md` for the full folder layout. Paths below are relative to this directory.

File	What it is
`qc_eigenmode.csv` (top level)	The full 42-point Qc(coupling_space, coupling_length) map, now with a `ceiling_flag` column (`ok`/`borderline`/`likely_saturated`) — current best data product, but **do not use `Qc` for `likely_saturated` rows**
`eigenmode_sweep/qc_eigenmode_vs_driven.csv`	The 24-point eigenmode-vs-driven-fit cross-check
`images/qc_eigenmode.png`, `images/qc_eigenmode_vs_driven.png`	thesis_style plots of the above
`driven_sweep/qc_prod_refit.csv`	The driven-sweep reference numbers (26 trustworthy + 9 edge-flagged + 4 noise-flagged)
`eigenmode_sweep/run_qc_eigenmode.py`	Production eigenmode Qc script (the recommended method going forward)
`driven_sweep/run_qc_sim.py`	Production driven-S21 script (recipe validated, but hits the shallow-notch ceiling on this design)
`isolation_tests/run_bareres_eigenmode_isolation.py`	The clean single-port analytic validation test
`../ansys_impedance_handoff/`	Original HFSS impedance cross-check (Zin mystery)
`ansys/eigenmode_qc_results/`	HFSS lossy-eigenmode cross-check — **done** (all 3 points), negative-but-well-diagnosed result: HFSS's own eigenmode also can't resolve the coupling Qc, for a mesh-adaptivity reason distinct from PALACE's noise-floor issue. See `SUMMARY.md`/`RESUME.md` there.
`../ansys_eigenmode_s21_crosscheck/`	Bare-resonator HFSS driven-S11 cross-check — **done**, see `ansys/ansys_results_s21/SUMMARY.md`
`isolation_tests/hanger_isolation_design.py`, `isolation_tests/run_hanger_isolation_eigenmode.py`	Two-port hanger isolation test (lumped-capacitor coupling) — raw Qc numbers **invalid** (see the "Overnight follow-up" section's `create_port_2_conds` finding); qualitative loaded_Q-vs-port-Q.csv agreement conclusion still stands
`isolation_tests/shuntc_validation_design.py`, `isolation_tests/run_shuntc_validation.py`	Resonance-free validation confirming `create_port_2_conds`'s capacitor is unreliable (~2-3× off even best-conditioned)

File	What it is
`eigenmode_sweep/flag_eigenmode_ceiling.py`	Flags `qc_eigenmode.csv` rows above the confirmed Qc~85,000 numerical-saturation ceiling
`analysis/plot_qc_length_fits.py`, `images/qc_length_fits.png`	The two analytic-model fit attempts (both fail) plotted against the trustworthy points
`images/`	Every figure referenced in this document

Bottom line

Use `qc_eigenmode.csv` as the working Qc(coupling_space, coupling_length) map, respecting its `ceiling_flag` column: **29/42 points (`ok`) are trustworthy**, backed by the ANSYS bare-resonator cross-check (1-2% agreement for `Qc` up to at least 2×10⁴, and the method holds up to a precisely-located ceiling at `Qc≈82,000-85,000` found overnight). **11/42 points (`likely_saturated`) and 2 more (`borderline`) — 26-31% of the grid — currently have no trustworthy Qc at all**: the eigenmode method saturates on a hard numerical noise floor above the ceiling, and the driven-S21 fit was never reliable in that regime either. If the actual chip design needs one of those (`cs22.5_cl150`, `cs25_cl300`, `cs15_cl150`, `cs10_cl100/150/250`, `cs25_cl100/150/250`, `cs22.5_cl200/250`, or the 2 borderline points), that Qc is currently unknown, not just uncertain. The two-port hanger isolation test's qualitative support for the mode-hybridization hypothesis stands, but its `Qc` numbers are invalid — `create_port_2_conds`'s capacitor was confirmed overnight to be uncalibrated (off by ~2-3× even in its best-conditioned test). `EIGENMODE_QC_ REQUEST.md` came back **negative but conclusive**: HFSS's own eigenmode solver independently fails to resolve Qc at these same design points too, for a different (mesh-adaptivity) reason than PALACE's. That convergence across two independent tools and four method/tool combinations is itself the strongest evidence yet that the 11 `likely_saturated` points sit in a coupling regime neither tool can currently characterize — not that any specific tool is broken. The most promising untried next step is PALACE eigenmode + spatial AMR targeted at the coupler gap, using our cluster's much larger RAM budget than what stopped HFSS's equivalent attempt.